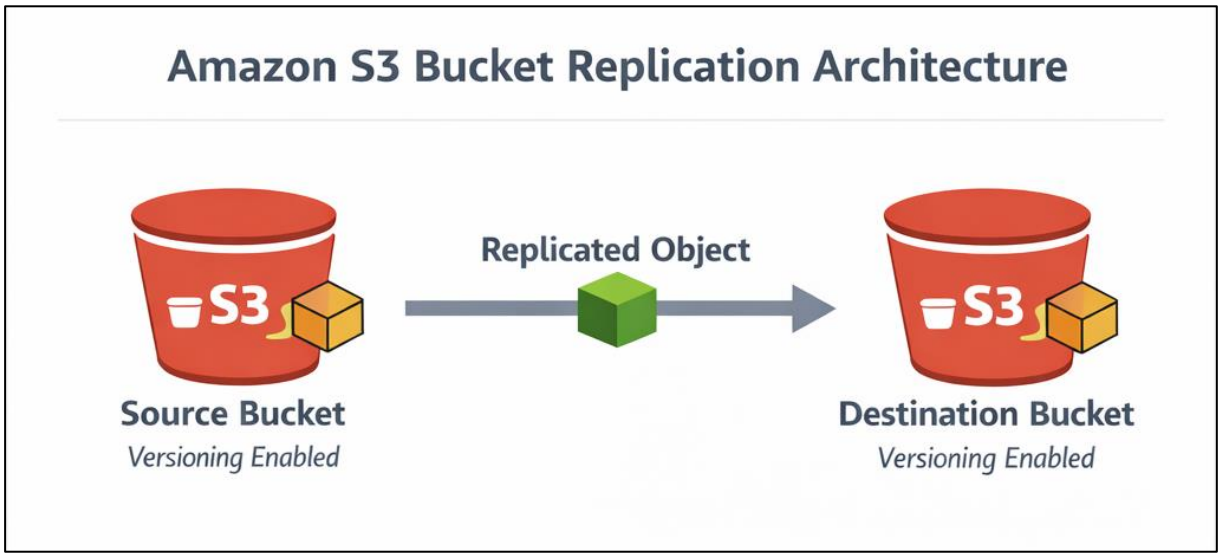
BUCKET REPLICATION IN S3

S3 Bucket Replication is a feature that automatically copies objects from a source bucket to a destination bucket, either within the same region or across different regions, to ensure data availability and durability. Bucket versioning must always be enabled on both source and destination buckets for replication to work.

Why S3 Bucket Replication is Used

- To protect data from regional failures
- To maintain backup and disaster recovery
- To meet compliance and regulatory requirements
- To provide low-latency access to users in different regions
- To ensure automatic synchronization of data between buckets
- Requires bucket versioning to be enabled on both buckets



Advantages of S3 Bucket Replication

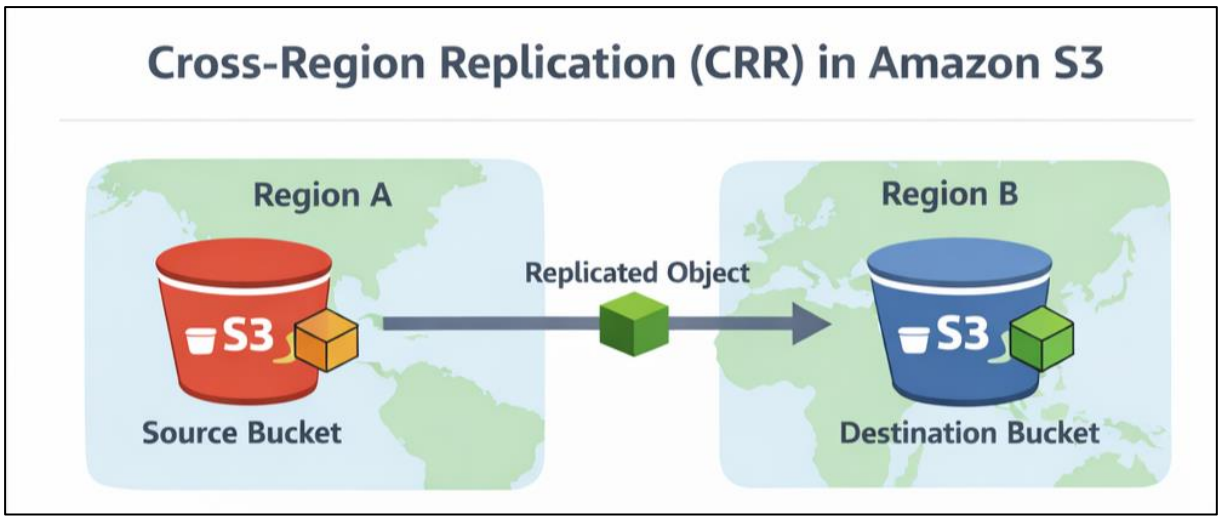
- Automatically replicates objects without manual intervention
- Improves data durability and high availability
- Supports Cross-Region Replication (CRR) and Same-Region Replication (SRR)
- Helps in business continuity and disaster recovery planning
- Ensures data consistency across buckets
- Versioning ensures safe replication of object updates and deletions

TYPES OF S3 BUCKET REPLICATION

1.Cross-Region Replication (CRR)

CRR automatically replicates objects from a source bucket to a destination bucket in a different AWS region.

- Disaster recovery
- Data protection across regions
- Global data access



2.Same-Region Replication (SRR)

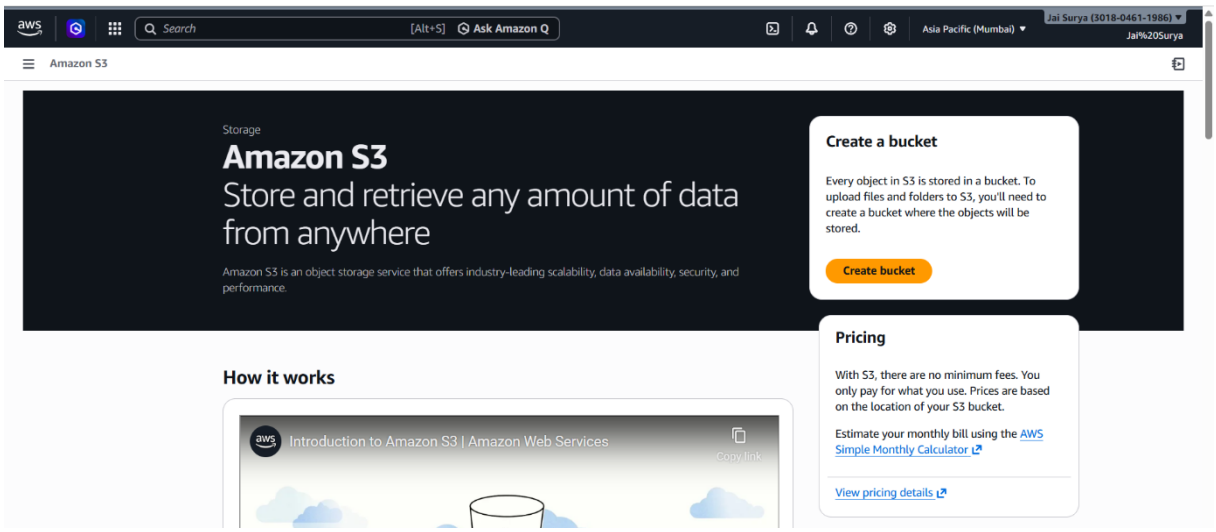
SRR replicates objects from a source bucket to a destination bucket **within the same AWS region**.

- Data aggregation
- Log replication
- Compliance requirements

HANDS-ON FOR BUCKET REPLICATION

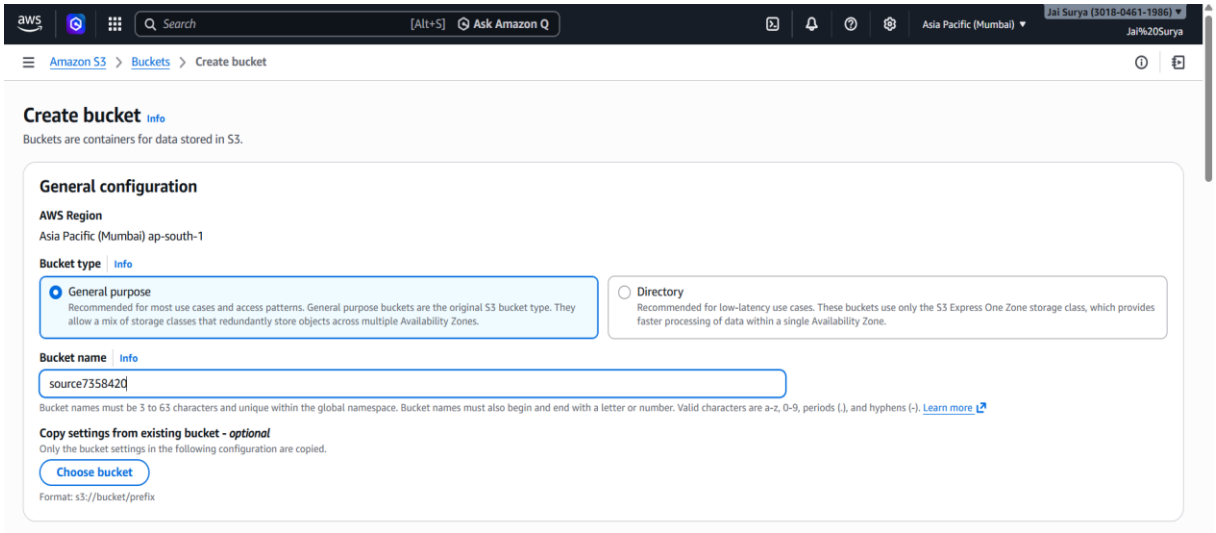
Screenshot 1: Create Bucket Page

1. Sign in to the AWS Management Console
2. Navigate to Amazon S3
3. Click Create bucket
4. The Create bucket configuration page is displayed



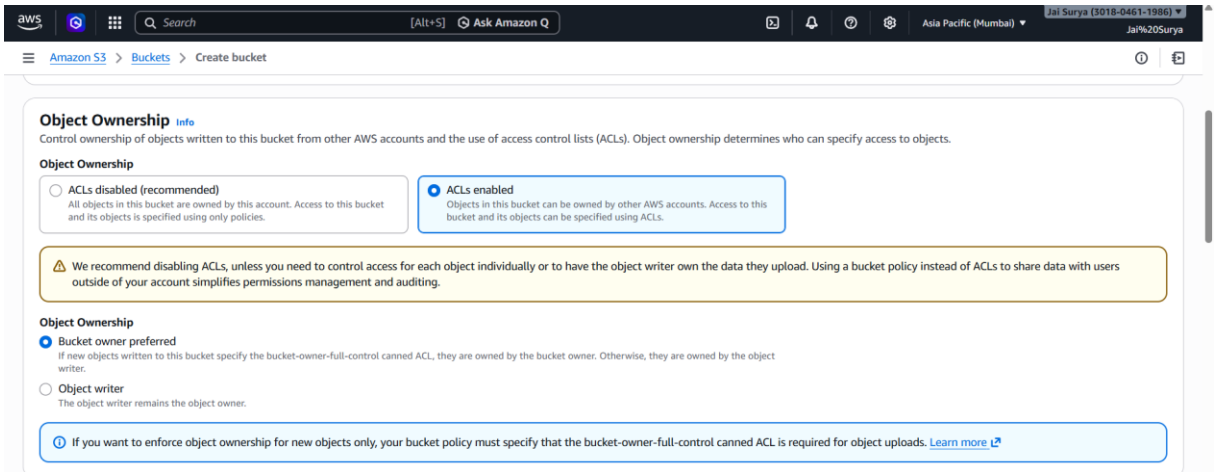
Screenshot 2: Select Bucket Type and Bucket Name

1. Select Bucket type as General purpose
2. Enter the bucket name as sourcebucket



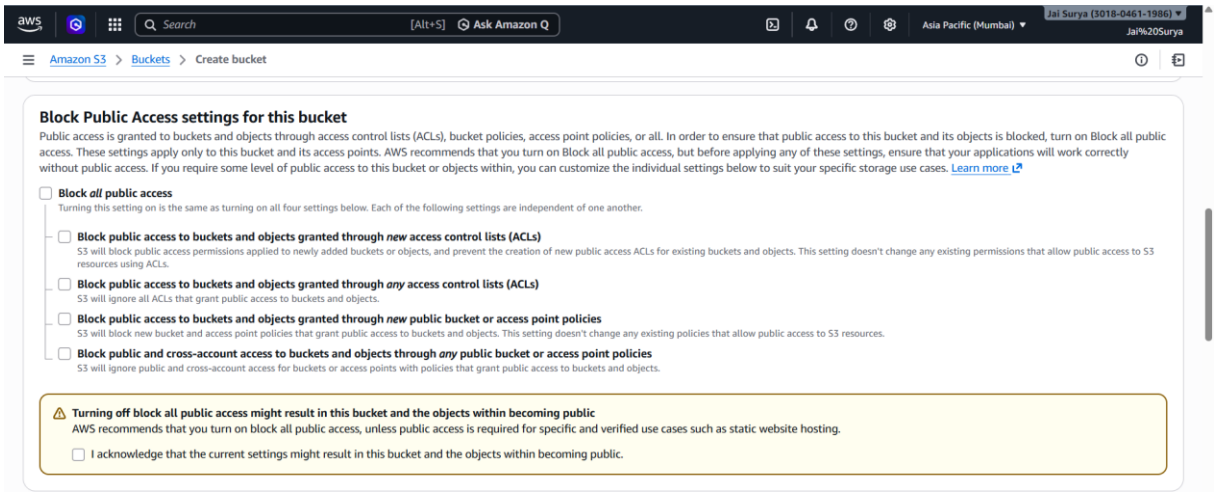
Screenshot 3: Object Ownership – Enable ACL (Public Bucket Setup)

1. Scroll to the Object Ownership section
2. Select ACLs enabled



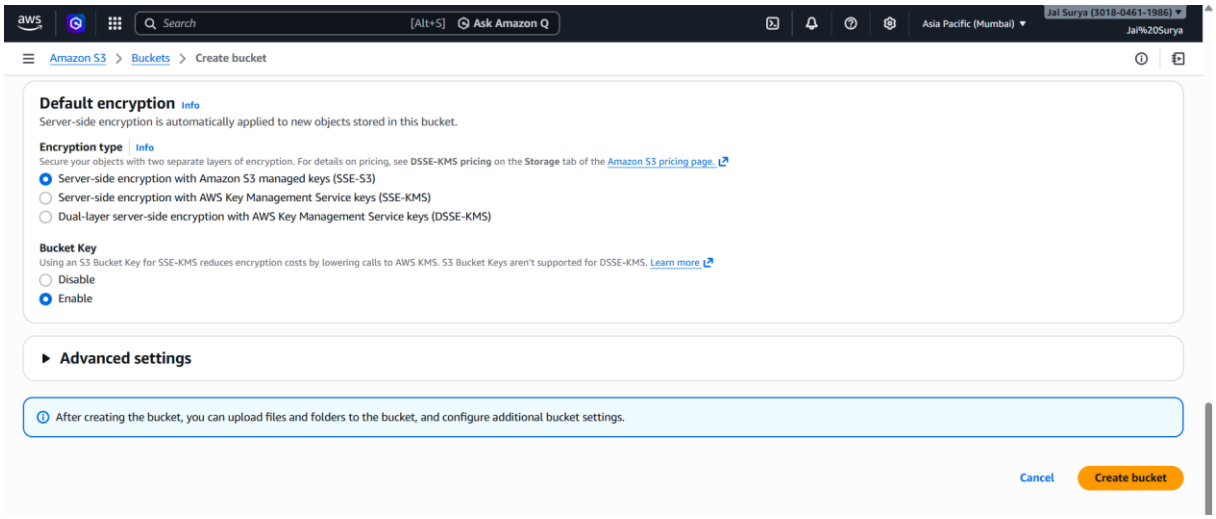
Screenshot 4: Disable Block Public Access

1. Scroll to the Block Public Access section
2. Uncheck the option Block all public access
3. Acknowledge the warning message
4. This allows the bucket to be configured as public



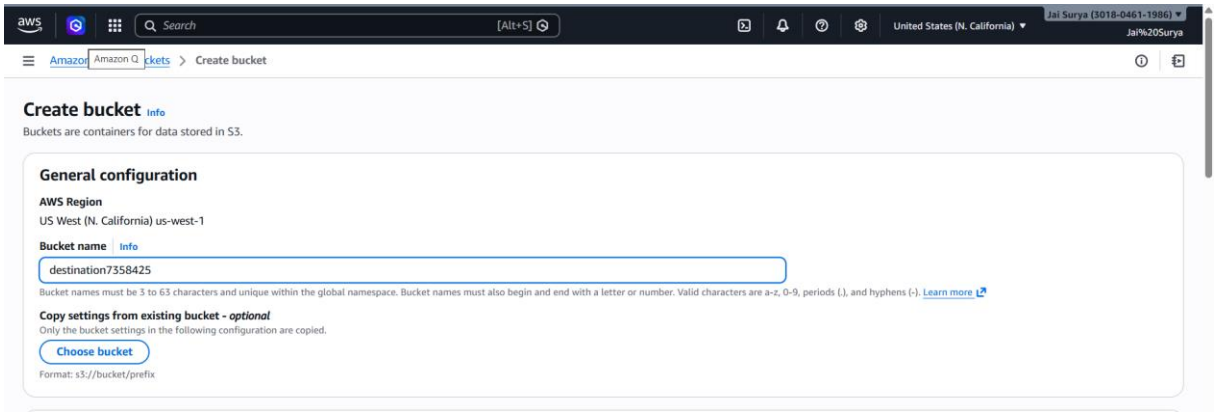
Screenshot 5: Encryption Settings and Bucket Creation

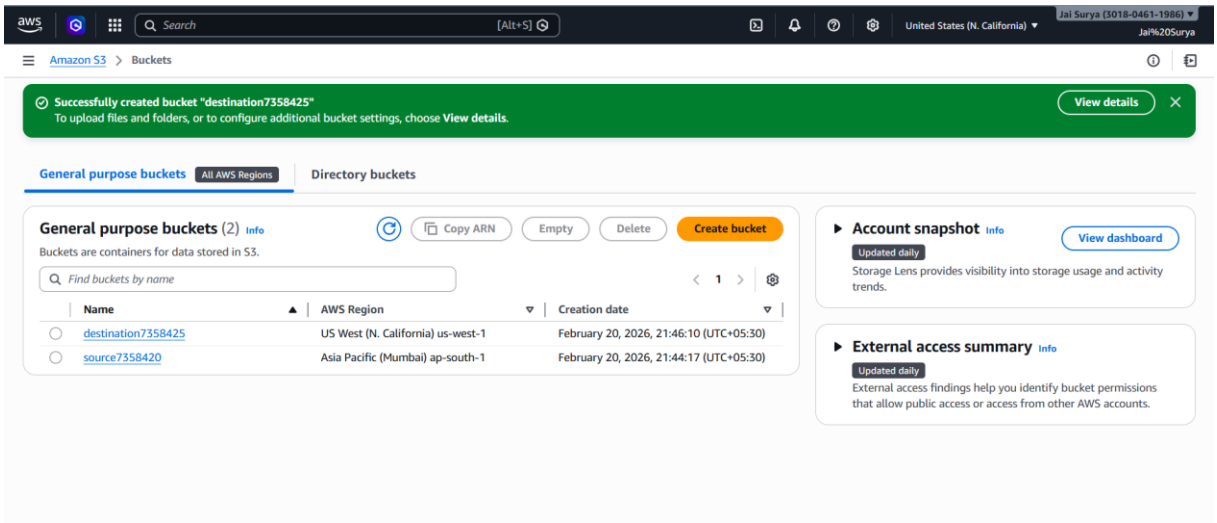
1. Scroll to the Default encryption section
2. Select encryption type as SSE-S3
3. Enable Bucket Key
4. Click Create bucket
5. The S3 bucket is created successfully



Screenshot 6 & 7: Destination Bucket Created Successfully

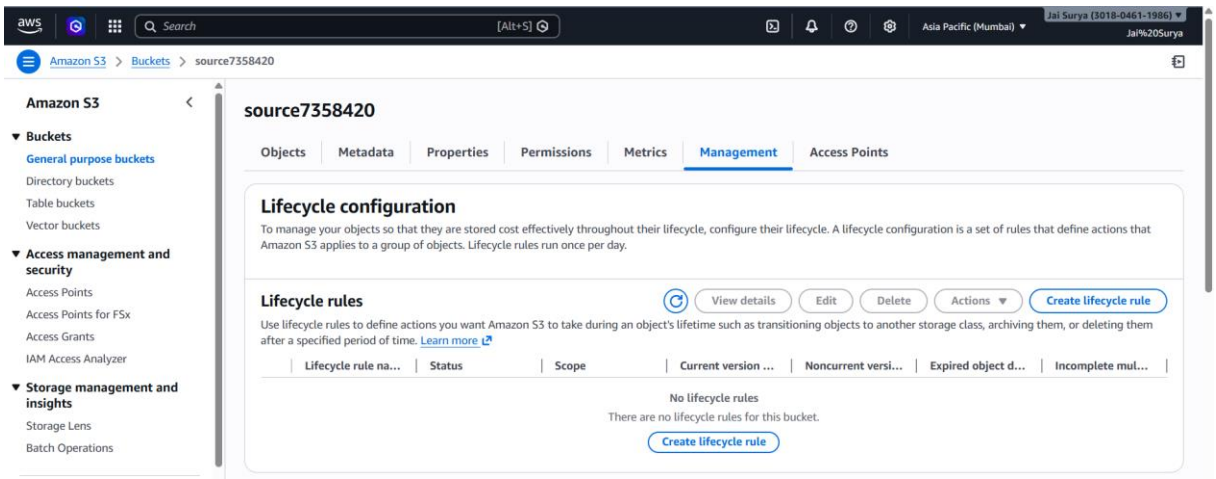
1. Repeat the bucket creation steps
2. Select General purpose as the bucket type
3. Enter a unique name for the destination bucket
4. Configure required settings (ACL, public access, encryption, versioning)
5. Click Create bucket
6. Destination bucket is created successfully





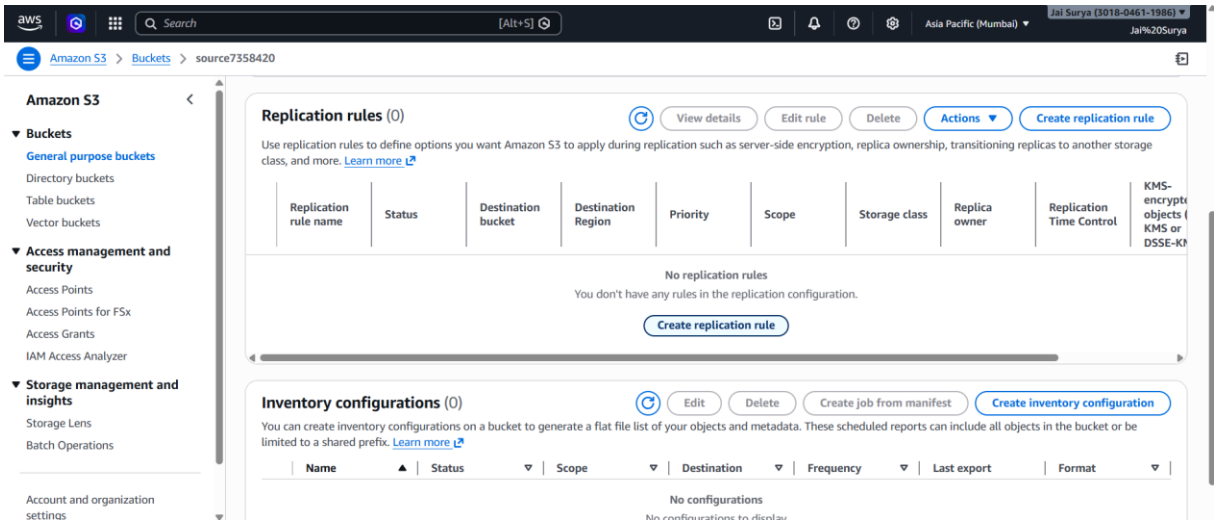
Screenshot 8: Source Bucket – Management Tab

1. Open the source S3 bucket
2. Navigate to the Management tab



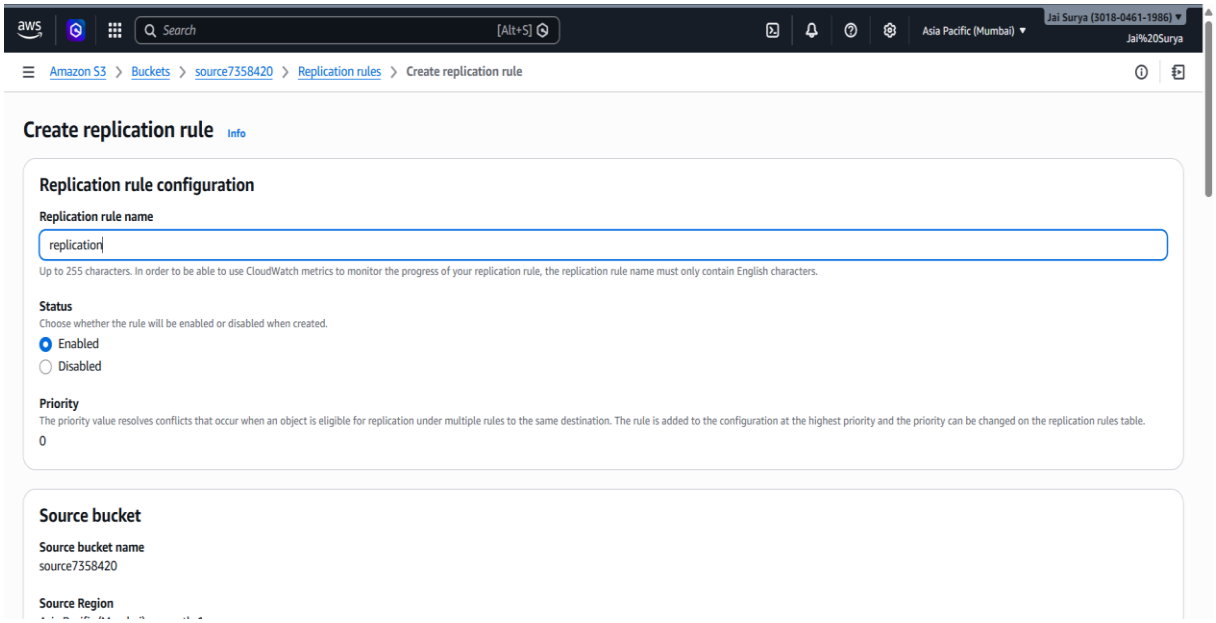
Screenshot 9: Replication Rules Section

1. In the Management tab, scroll to Replication rules
2. Click Create replication rule
3. Start configuring bucket replication settings



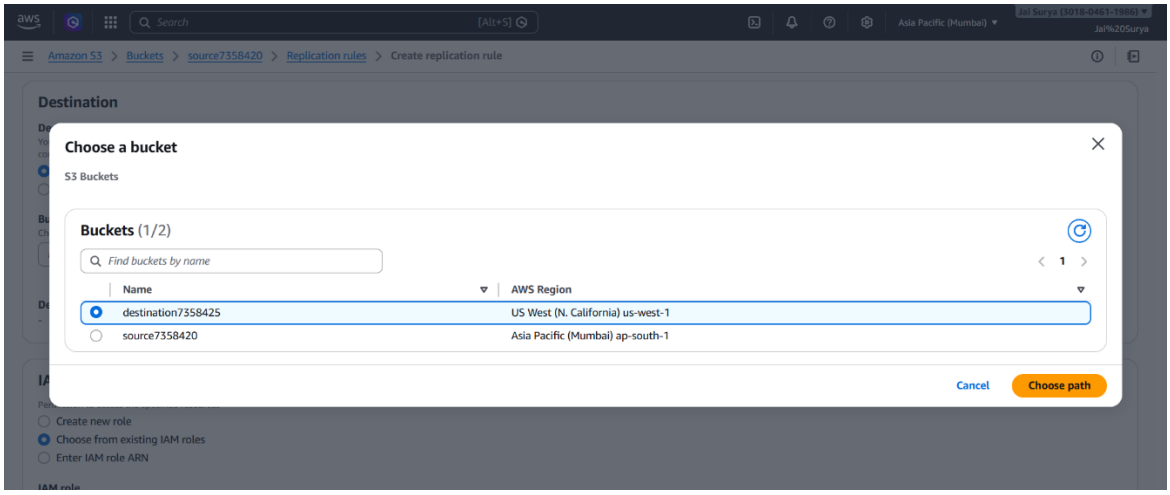
Screenshot 10: Configure Replication Rule Name and Status

1. Enter a Replication rule name
2. Set the Status to Enabled



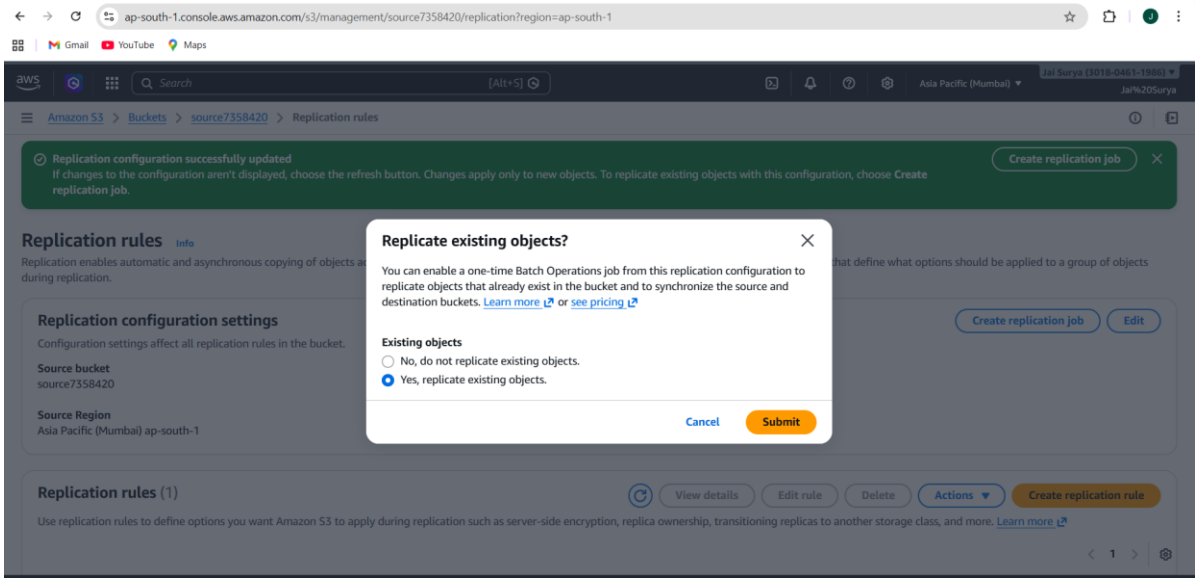
Screenshot 11: Choose Destination Bucket

1. In the replication rule settings, locate Destination
2. Select Choose a bucket in this account
3. Choose the destination bucket from the list
4. Specify the destination path (same or different prefix if required)
5. Proceed to the next configuration step



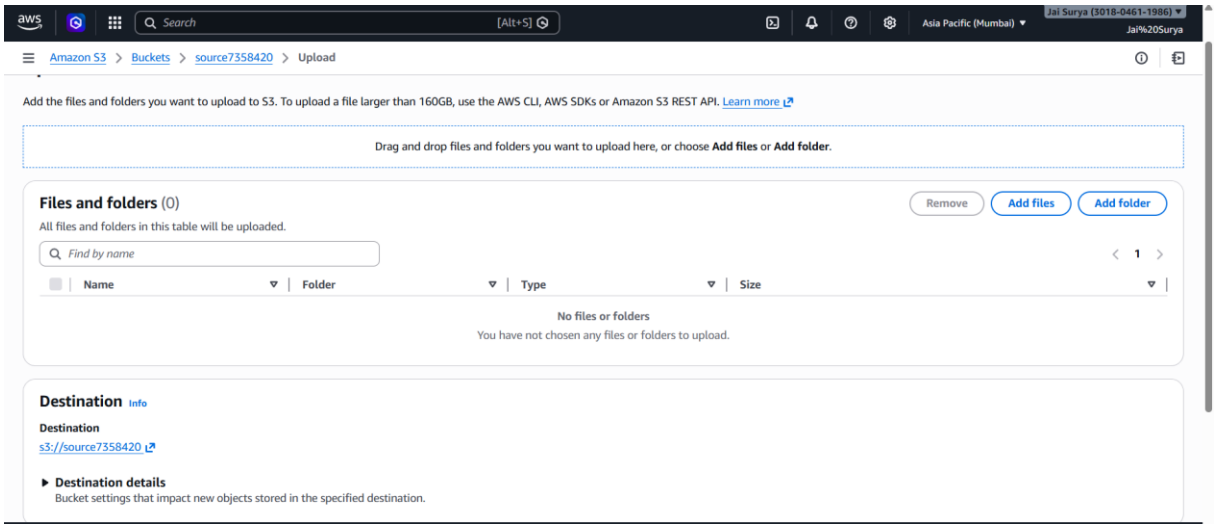
Screenshot 12: Existing Object Replication

1. In the replication rule settings, locate Existing object replication
2. Choose Enable to replicate existing objects



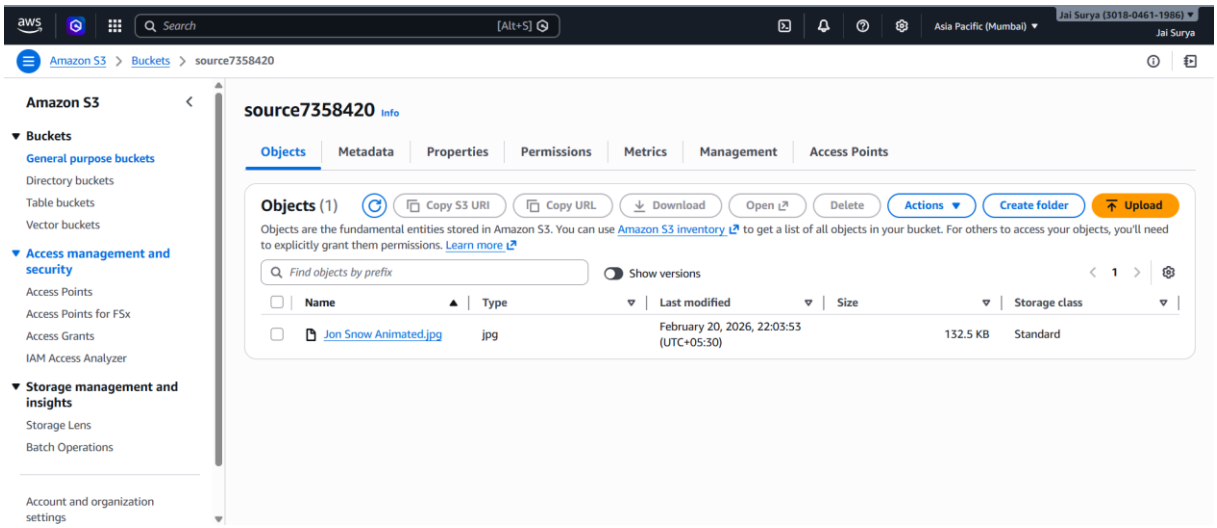
Screenshot 13: Upload Object to Source Bucket

1. Open the source S3 bucket
2. Click Upload
3. Choose a file from the local system
4. Click Upload to add the object to the source bucket



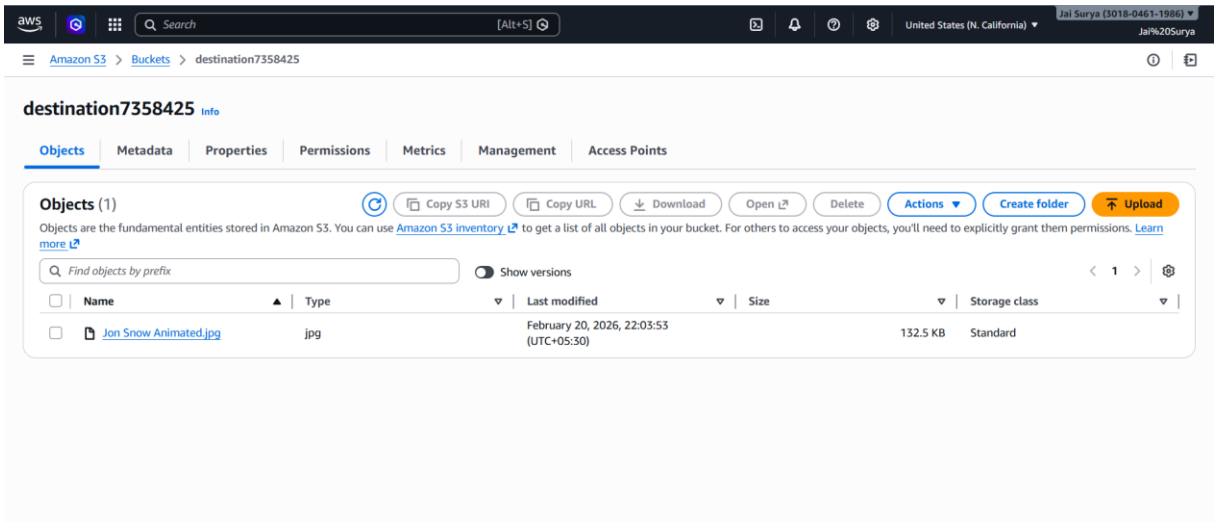
Screenshot 14: Object Uploaded Successfully

1. Verify that the file appears in the source bucket object list
2. Confirm the upload status shows successful
3. The object is now available for replication



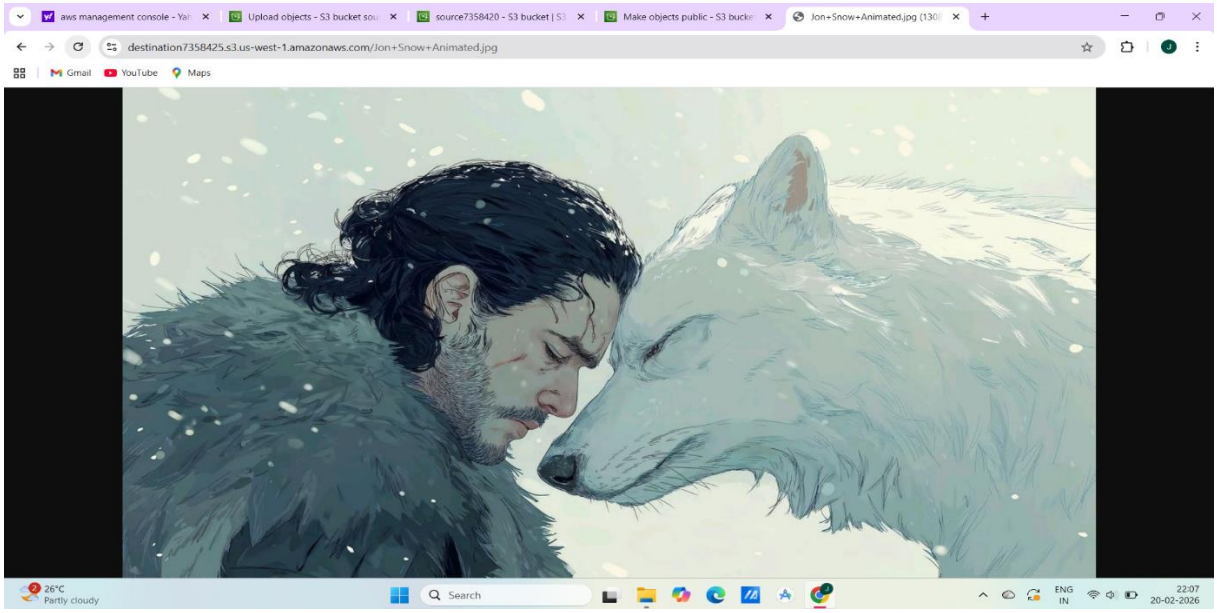
Screenshot 15: Replicated Object in Destination Bucket

1. Duplicate the browser tab or open a new tab
2. Navigate to the destination S3 bucket
3. Verify that the uploaded object appears automatically
4. This confirms that bucket replication is successful



Screenshot 16: Access Replicated Image Using Object URL

1. Open the destination S3 bucket
2. Select the replicated image object
3. Copy the Object URL
4. Paste the URL in a new browser tab
5. The image opens successfully, confirming replication and public access



STATIC WEBSITE HOSTING IN S3

Definition

Static Website Hosting is a feature of Amazon S3 that allows users to host static content such as HTML, CSS, JavaScript, and images directly from an S3 bucket without using a server. The content is accessed using a public endpoint URL provided by S3.

Why Static Website Hosting is Used

- To host simple websites without managing servers
- To reduce infrastructure and maintenance costs
- To deliver content quickly and reliably
- To host portfolio websites, landing pages, and documentation
- To integrate with CDN services for faster global access

Advantages of Static Website Hosting

- No server management required
- Highly scalable and cost-effective
- High availability and durability
- Easy deployment and maintenance

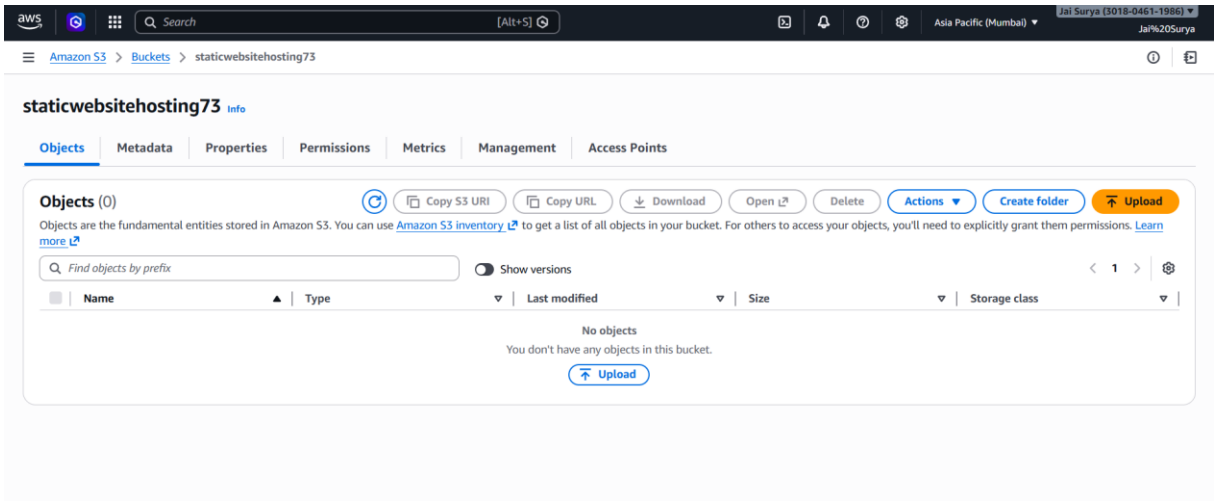
Key Points (Important for Exam & Assignment)

- Bucket must be public
- Static website hosting must be enabled in bucket properties
- Requires an index document (example: index.html)
- Error document is optional

HANDS-ON FOR STATIC WEBSITE HOSTING IN S3

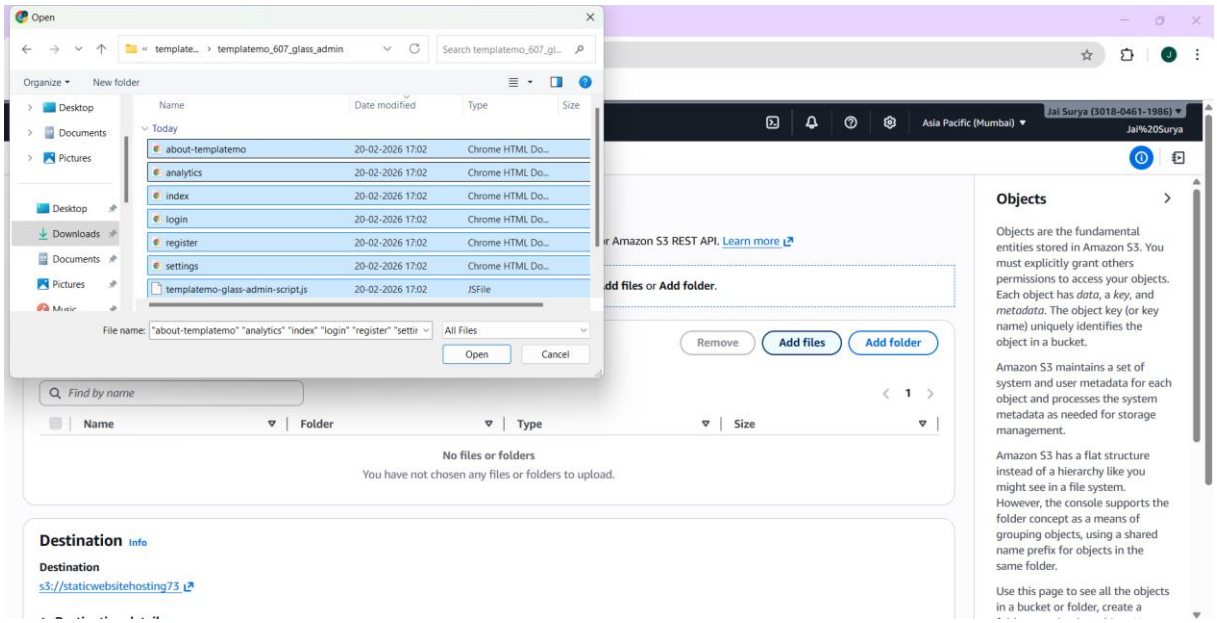
Screenshot 1: Create Bucket, Open Bucket, Click Upload

- Click Create bucket in the AWS S3 console.
- Enter a bucket name (e.g., staticwebsitehosting) and select the desired region.
- Click Create bucket.
- Open the newly created bucket.
- Click Upload to start adding website files (HTML, CSS, JS, images).



Screenshot 2: Add Files to Bucket

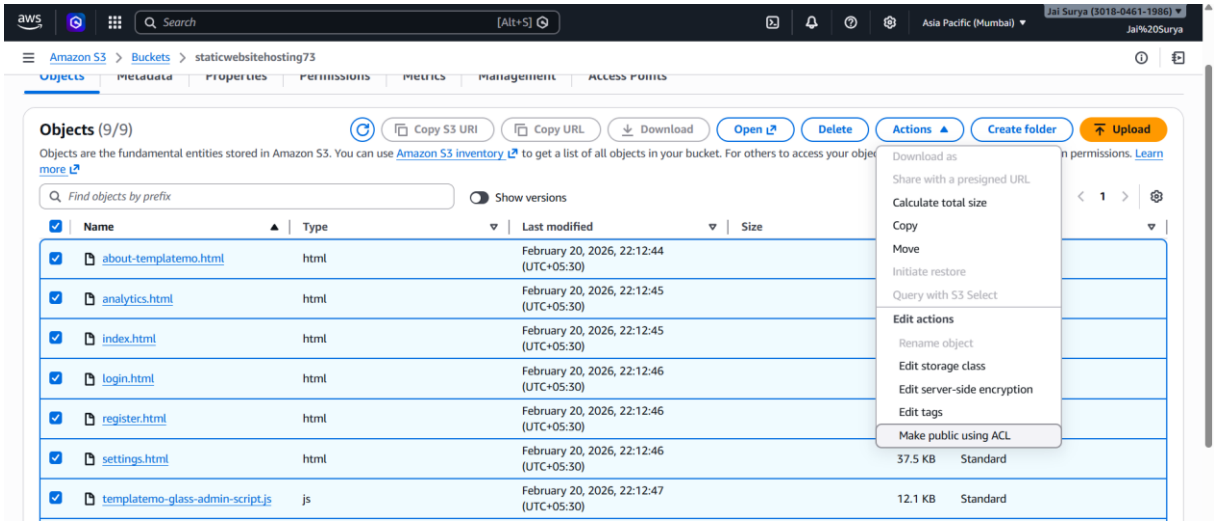
1. In the Upload window, click Add files.
2. Select your website files from the local system (e.g., index.html, CSS, JS, images).
3. Confirm the selected files appear in the upload list.
4. Click Next to proceed with configuration options.



Screenshot 3: Make Objects Public Using ACL

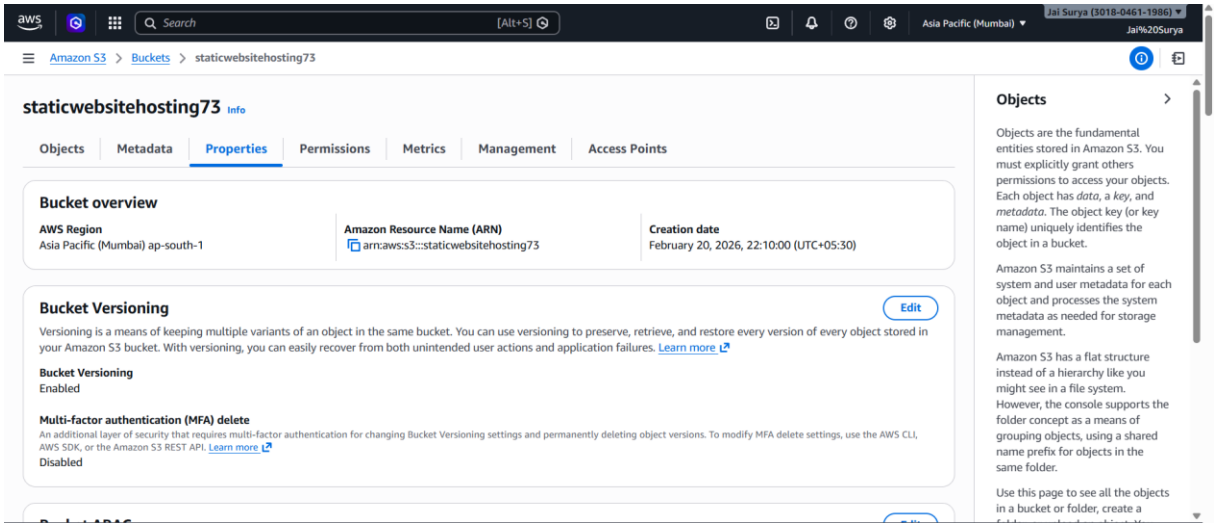
1. After uploading files, select all the objects in the bucket.
2. Click on Actions → Edit access control list (ACL).
3. Enable Read access for Everyone (public access).

4. Save changes to apply public access to all objects.
5. This allows the website files to be publicly accessible via their URLs.



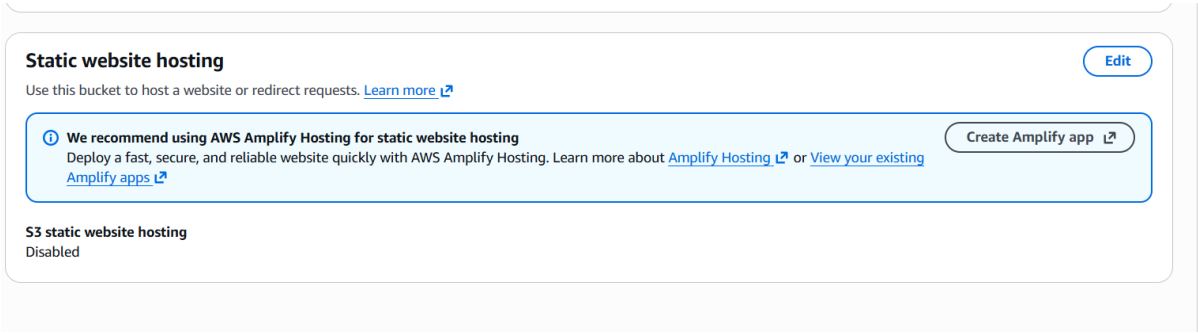
Screenshot 4: Enable Versioning in Bucket Properties

1. Open the bucket in the AWS S3 console.
2. Go to the Properties tab.
3. Scroll to the Bucket Versioning section.
4. Click Enable versioning.
5. Save changes to activate versioning for the bucket.



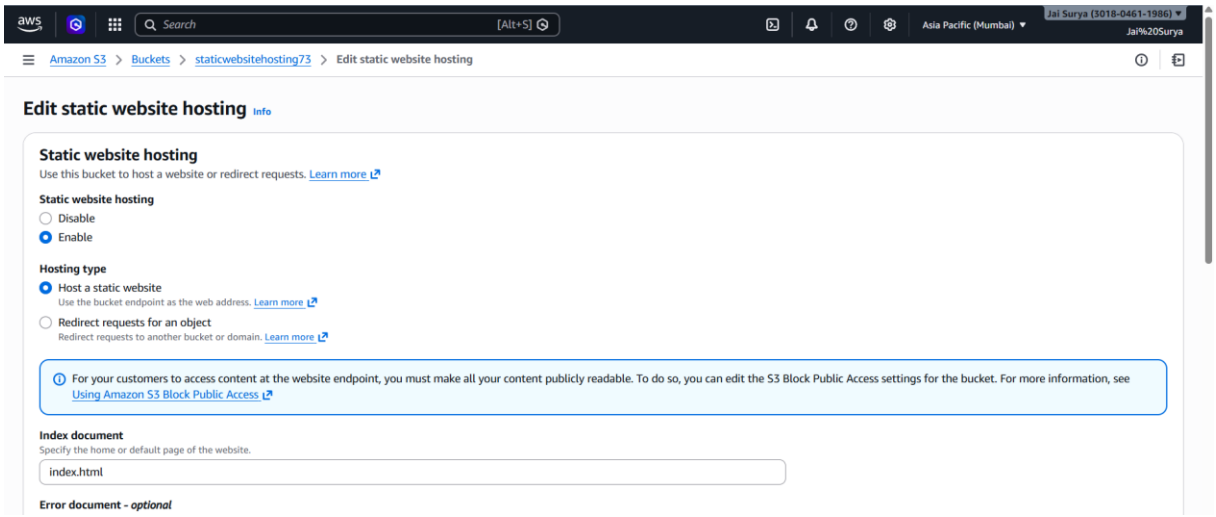
Screenshot 5: Enable Static Website Hosting

1. In the bucket Properties tab, scroll down to **Static website hosting**.
2. Click Edit to configure website hosting settings.



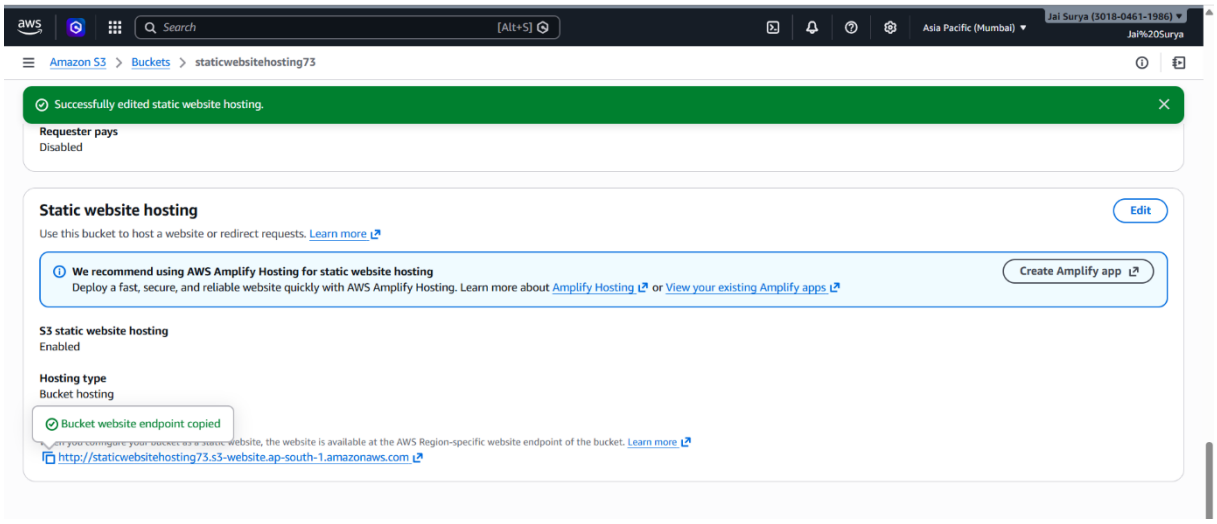
Screenshot 6: Configure Index Document for Static Website

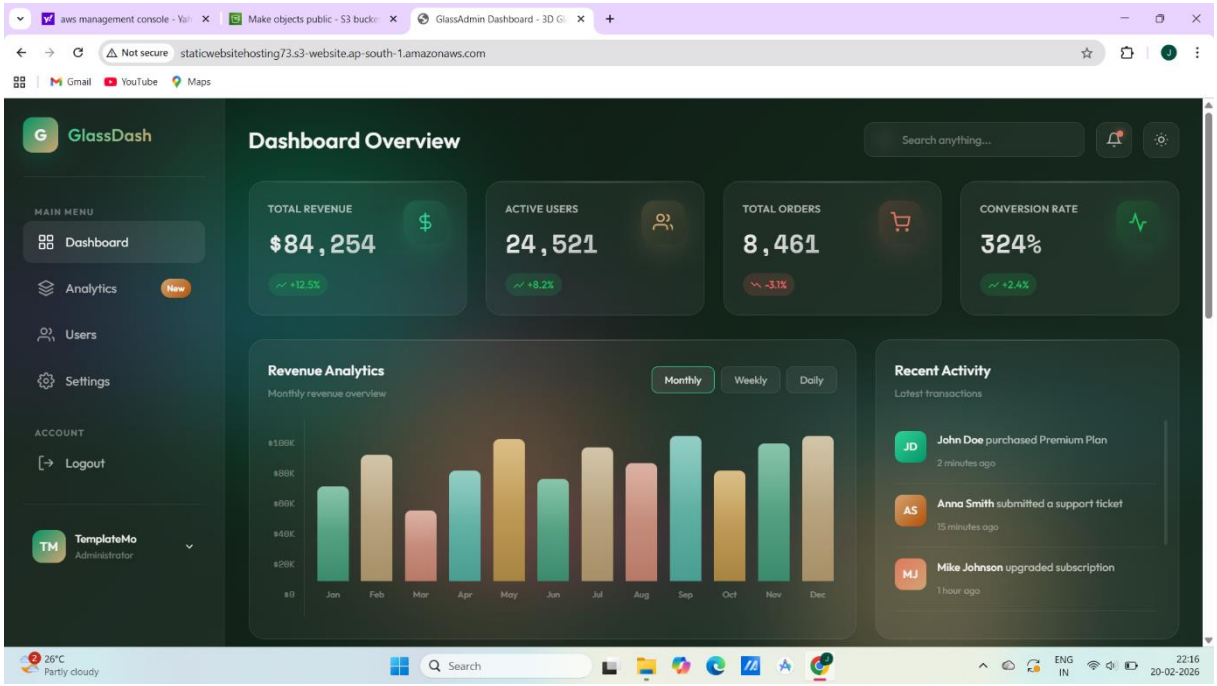
1. In the Static Website Hosting section, select Enable.
2. Enter the Index document name (e.g., index.html).
3. Click Save changes to enable the bucket as a static website.



Screenshot 7: Copy Static Website Endpoint URL

1. In the Static Website Hosting section of the bucket, locate the Bucket website endpoint.
2. Click the Copy URL button.
3. The URL can be used to access your static website publicly.
4. This URL points directly to the hosted website on Amazon S3



Screenshot 8: Open Static Website Using Endpoint URL

1. Open a new browser tab.
2. Paste the copied static website endpoint URL into the address bar.
3. Press Enter.
4. The static website loads successfully, confirming that the bucket is correctly configured for hosting.